

Sleep Project Final Report

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1. Introduction

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2. Methods

(paste methods here)

Step by Step Explanation (add if wanted):

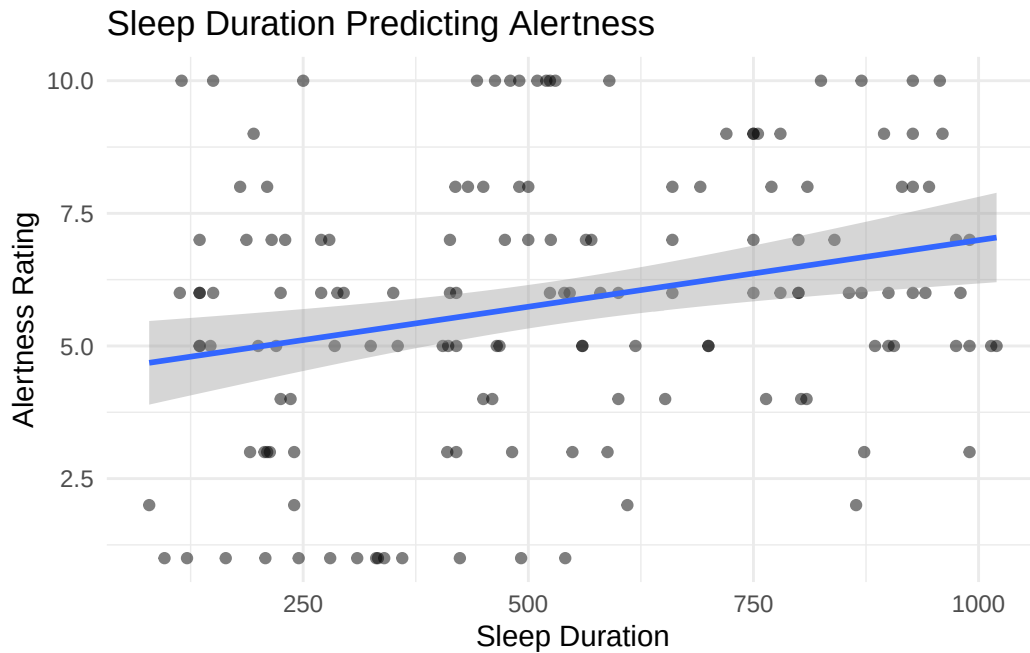
- **Set up** *(add if wanted)*

3. Results

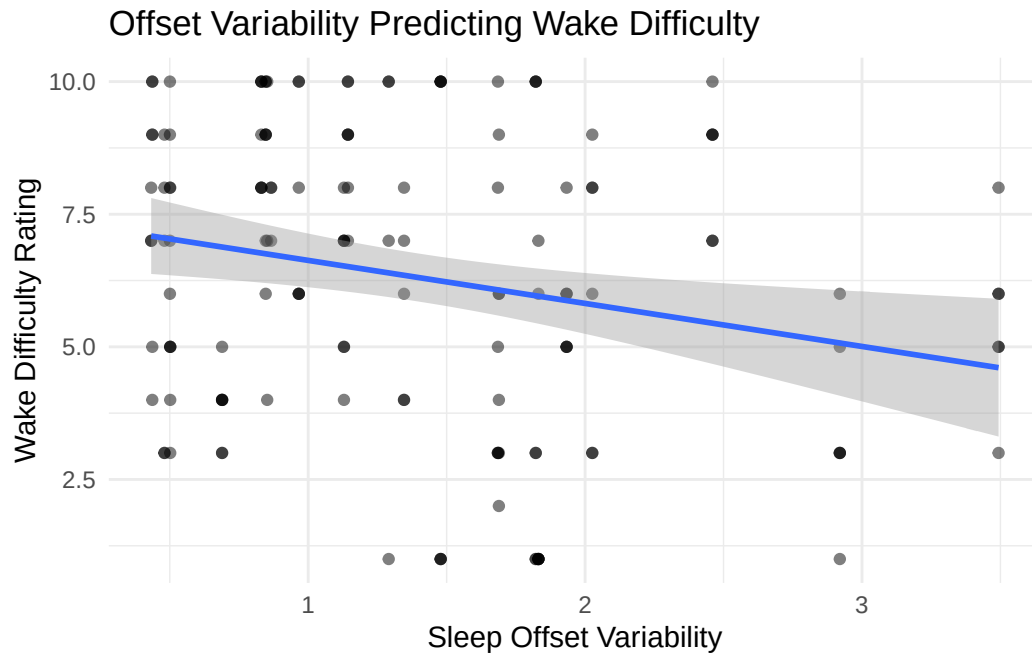
- **Descriptive Results**
 - Through our upper quartile study on sleep quality we discovered that the top-quartile sleepers had a much longer sleep duration as well as lower caffeine use and less device time compared to sleepers who were not in the top-quartile. Unexpectedly, top sleepers also had higher anxiety ratings, more social media use, and less physical activity.

- **Regression Results**

- The regression study was based on the independent variables: onset time, offset time, onset variability, offset variability, sleep duration and the dependent variables: alertness, anxiety, wake difficulty, and sleep quality. Due to the recurring problem of heteroscedasticity violating the assumption of homoscedasticity in the linear regression analyses conducted for each dependent variable we used an HC3 heteroskedasticity-robust standard errors test to interpret coefficients more accurately. The results showed that sleep duration is a significant predictor for alertness ($p= 0.0096$, $\beta = 0.0023$), indicating that longer sleep duration is associated with higher alertness.

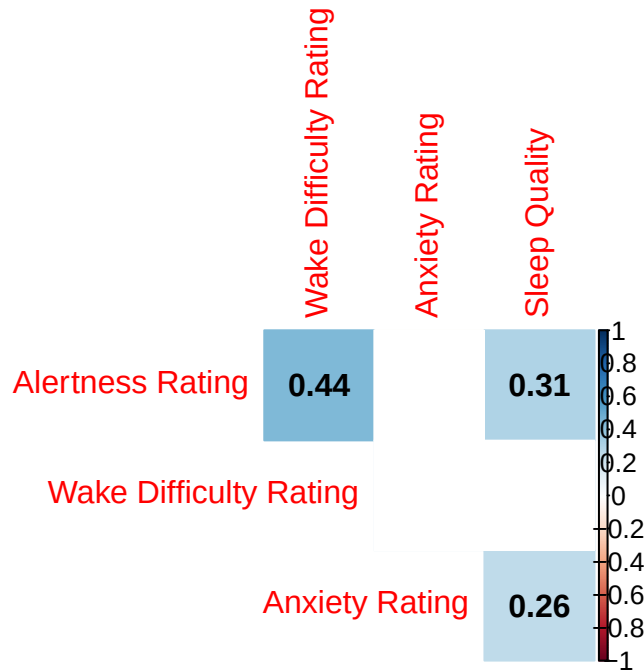


- Another notable result is offset variability as a significant predictor for wake difficulty.



- **Correlation Results**

- We also looked into correlation between the dependent variables and found significant correlations between alertness & wake difficulty ($r=0.42$), alertness & sleep quality ($r=0.31$), and anxiety & sleep quality ($r=0.30$).



- **Mixed Effects Results**

- The data included 147 nights from 27 participants, within the data we noticed some trends although not statistically significance still peaked our interest. The trends we found were the association of more sleep than usual to a better reported sleep quality as well as a later sleep onset associated with a worse reported sleep quality. Though these are interesting trends it's important to emphasize that no predictors reached statistical significance in this study. The model assumptions were mostly satisfied showing no major variance issues and although there were some slight left-skew residuals the overall model fit is considered acceptable.

- **Takeaways**

- Sleep duration is the strongest consistent predictor within our study. However, most other sleep variables showed limited predictive powers. Some relationships exist between key outcomes, but casual predictors are weak overall.

4. Conclusions/Future Work

(paste conclusion here)

5. Appendix

Call:

```
lm(formula = `Alertness Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.6906	-1.5393	0.0393	1.6834	5.1780

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.7175013	1.2809727	2.902	0.00431 **
`Sleep Duration`	0.0022772	0.0008369	2.721	0.00734 **
`Sleep Offset Time_Decimal Hour`	0.0353955	0.1320510	0.268	0.78906
`Sleep Onset Decimal Hour`	0.0382507	0.0239230	1.599	0.11211
onset_var	0.0585856	0.0449249	1.304	0.19436
offset_var	-0.2821057	0.2730019	-1.033	0.30324

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.486 on 139 degrees of freedom

(27 observations deleted due to missingness)

Multiple R-squared: 0.1135, Adjusted R-squared: 0.08165

F-statistic: 3.56 on 5 and 139 DF, p-value: 0.004636

Call:

```
lm(formula = `Wake Difficulty Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +  
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.3187	-2.1928	0.3314	2.0396	4.9466

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.992346	1.400665	3.564	0.000504 ***
`Sleep Duration`	0.001123	0.000929	1.209	0.228878
`Sleep Offset Time_Decimal Hour`	0.217096	0.144162	1.506	0.134408
`Sleep Onset Decimal Hour`	0.031007	0.026295	1.179	0.240379

```
onset_var          -0.064145    0.049466   -1.297 0.196913
offset_var         -0.719649    0.300054   -2.398 0.017824 *
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.703 on 136 degrees of freedom

(30 observations deleted due to missingness)

Multiple R-squared: 0.1076, Adjusted R-squared: 0.07479

F-statistic: 3.28 on 5 and 136 DF, p-value: 0.007913

Call:

```
lm(formula = `Anxiety Rating` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.9946	-0.6792	1.0401	1.4167	2.1795

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.3598911	1.1990470	6.972	1.18e-10 ***
`Sleep Duration`	0.0011577	0.0007859	1.473	0.143
`Sleep Offset Time_Decimal Hour`	0.0354732	0.1231159	0.288	0.774
`Sleep Onset Decimal Hour`	-0.0248727	0.0223228	-1.114	0.267
onset_var	-0.0309295	0.0420798	-0.735	0.464
offset_var	-0.0512598	0.2551679	-0.201	0.841

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.318 on 138 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.02573, Adjusted R-squared: -0.009565

F-statistic: 0.729 on 5 and 138 DF, p-value: 0.6028

Call:

```
lm(formula = `Sleep Quality` ~ `Sleep Duration` + `Sleep Offset Time_Decimal Hour` +
  `Sleep Onset Decimal Hour` + onset_var + offset_var, data = step1)
```

Residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

-6.4510 -1.0031 0.3062 1.4986 3.0138

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.8921655	1.0638612	5.538	1.49e-07 ***
`Sleep Duration`	0.0002554	0.0006951	0.367	0.714
`Sleep Offset Time_Decimal Hour`	0.1343439	0.1099508	1.222	0.224
`Sleep Onset Decimal Hour`	0.0320627	0.0199638	1.606	0.111
onset_var	-0.0102492	0.0376939	-0.272	0.786
offset_var	-0.0367834	0.2268967	-0.162	0.871

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.064 on 138 degrees of freedom

(28 observations deleted due to missingness)

Multiple R-squared: 0.03469, Adjusted R-squared: -0.0002878

F-statistic: 0.9918 on 5 and 138 DF, p-value: 0.4252

studentized Breusch-Pagan test

data: alertness3

BP = 12.35, df = 5, p-value = 0.0303

studentized Breusch-Pagan test

data: wake_diff3

BP = 17.556, df = 5, p-value = 0.003558

studentized Breusch-Pagan test

data: anxiety3

BP = 10.383, df = 5, p-value = 0.06509

studentized Breusch-Pagan test

data: sleep_quality3

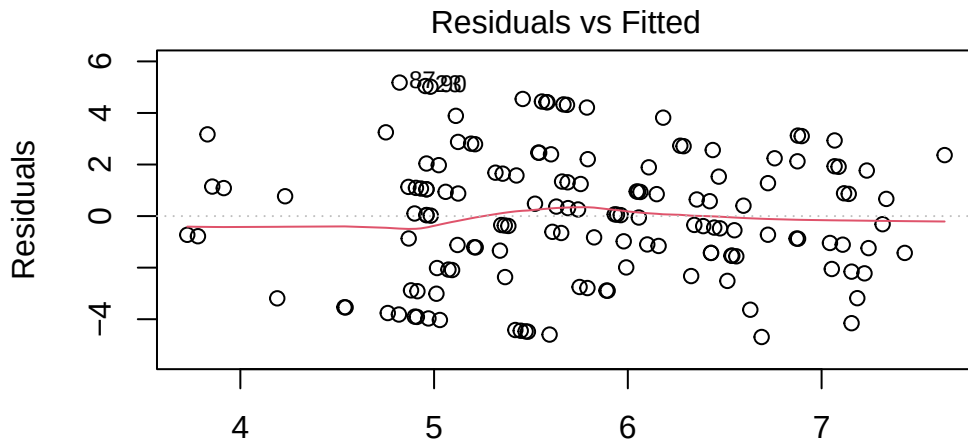
BP = 13.291, df = 5, p-value = 0.0208

`Sleep Duration`	`Sleep Offset Time_Decimal Hour`
1.204114	1.208512
`Sleep Onset Decimal Hour`	onset_var
1.376575	1.053563
offset_var	
1.078269	

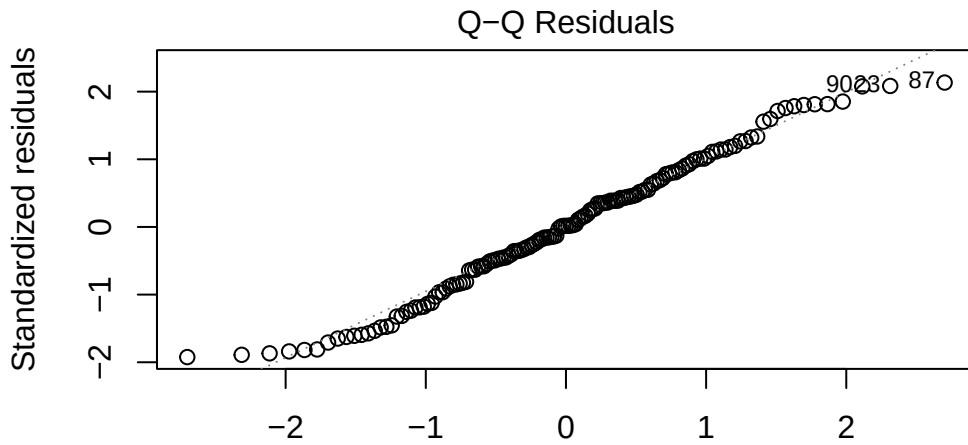
`Sleep Duration`	`Sleep Offset Time_Decimal Hour`
1.210630	1.216753
`Sleep Onset Decimal Hour`	onset_var
1.378852	1.045759
offset_var	
1.079007	

`Sleep Duration`	`Sleep Offset Time_Decimal Hour`
1.202285	1.205785
`Sleep Onset Decimal Hour`	onset_var
1.375963	1.049018
offset_var	
1.073044	

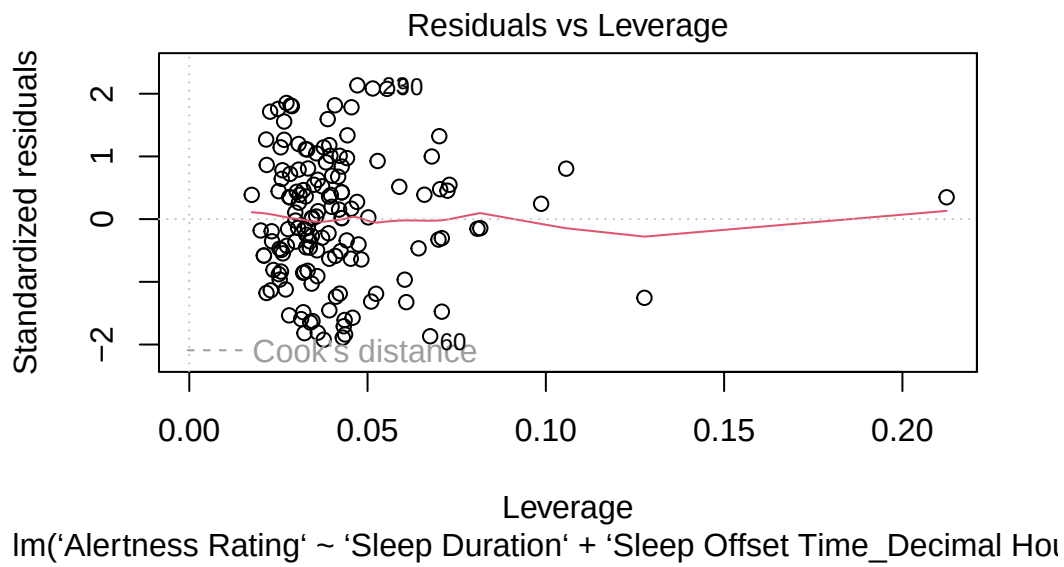
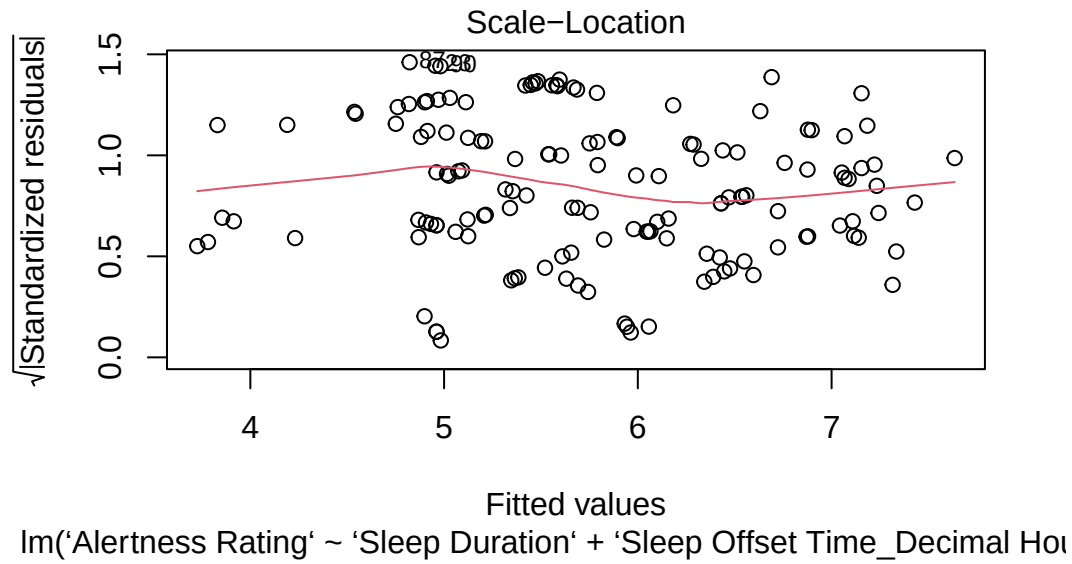
`Sleep Duration`	`Sleep Offset Time_Decimal Hour`
1.204022	1.203788
`Sleep Onset Decimal Hour`	onset_var
1.372275	1.062445
offset_var	
1.077400	

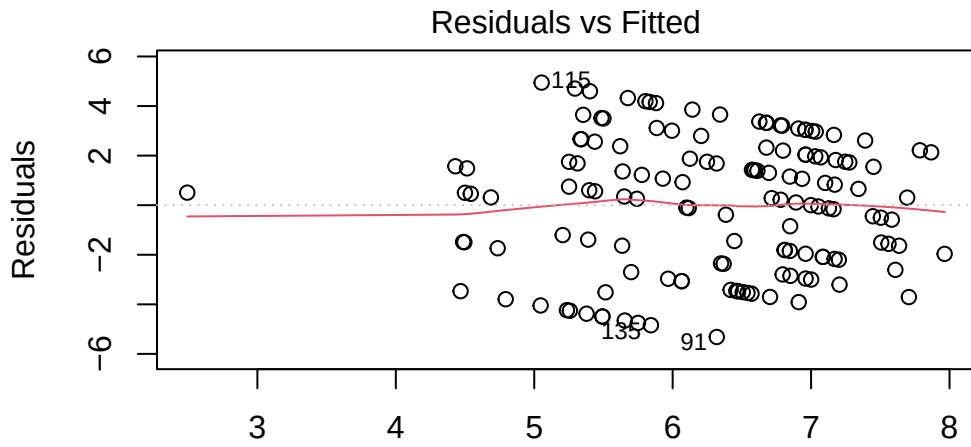


Fitted values
 $\text{lm}(\text{'Alertness Rating'} \sim \text{'Sleep Duration'} + \text{'Sleep Offset Time_Decimal Hours'})$

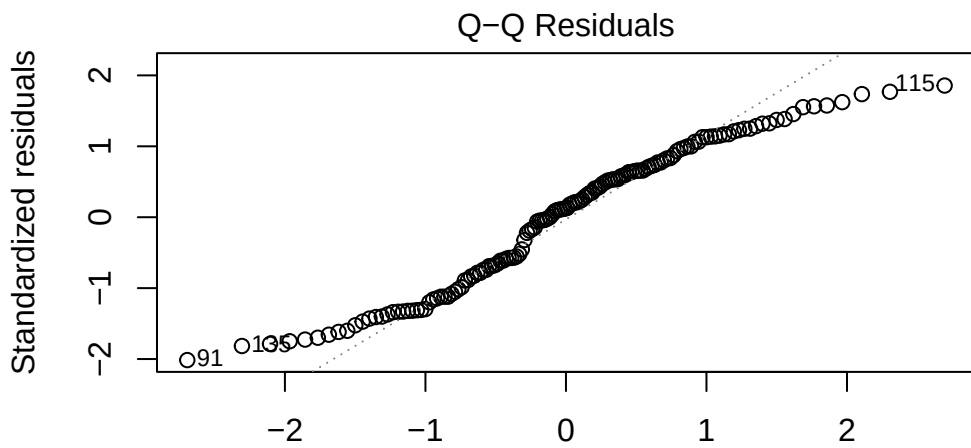


Theoretical Quantiles
 $\text{lm}(\text{'Alertness Rating'} \sim \text{'Sleep Duration'} + \text{'Sleep Offset Time_Decimal Hours'})$

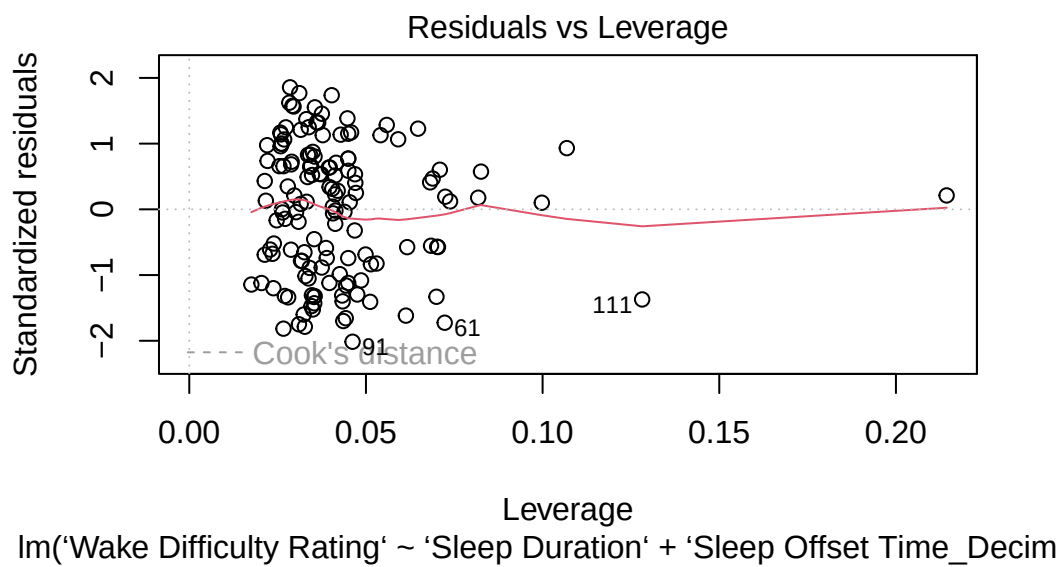
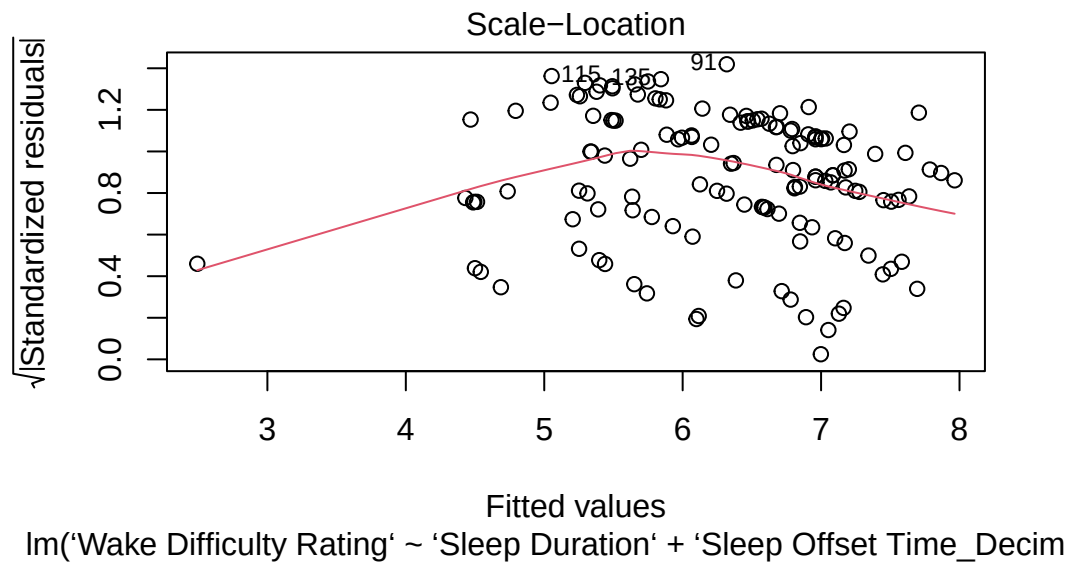


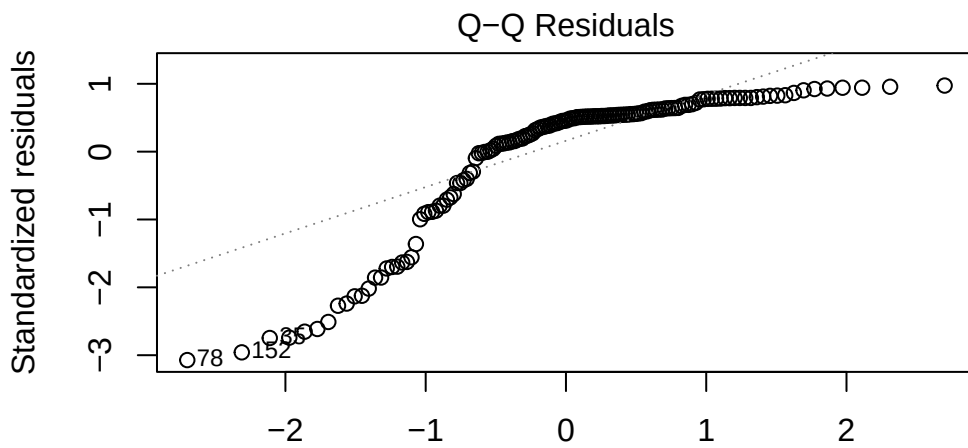
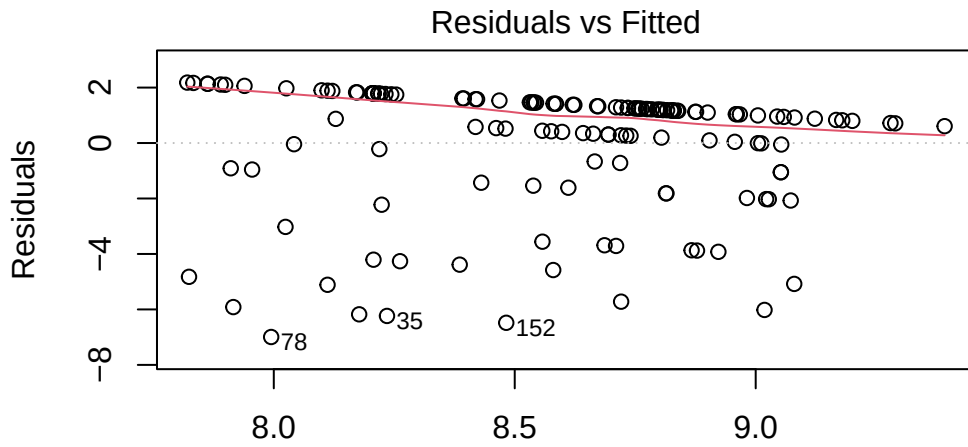


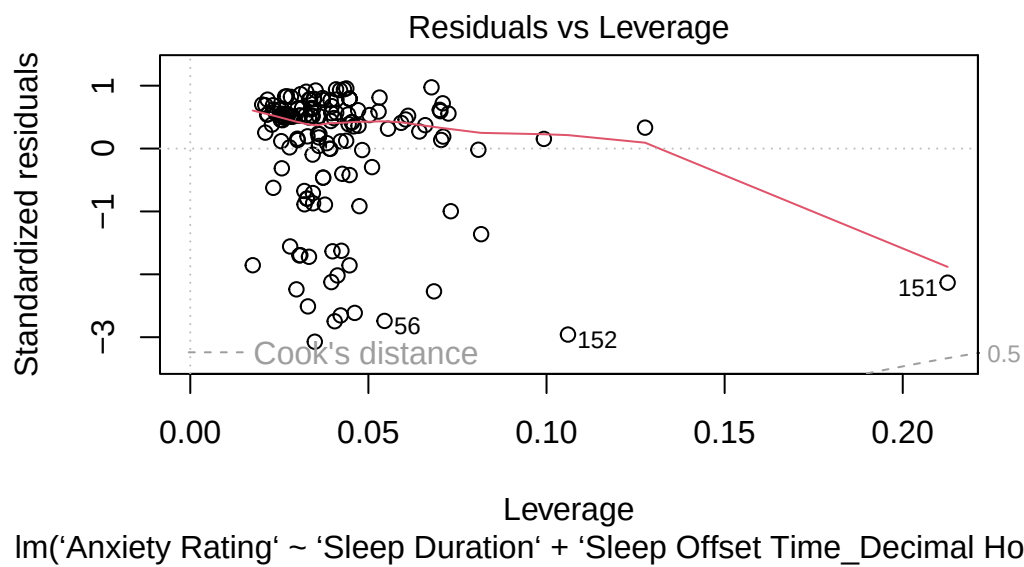
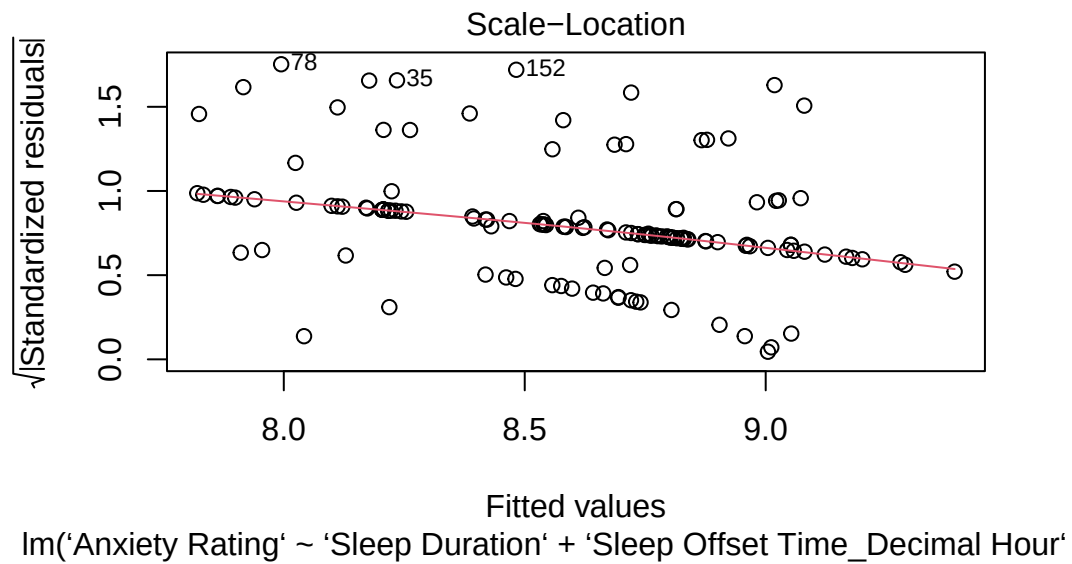
Fitted values
 $\text{lm}(\text{'Wake Difficulty Rating'} \sim \text{'Sleep Duration'} + \text{'Sleep Offset Time_Decim'})$

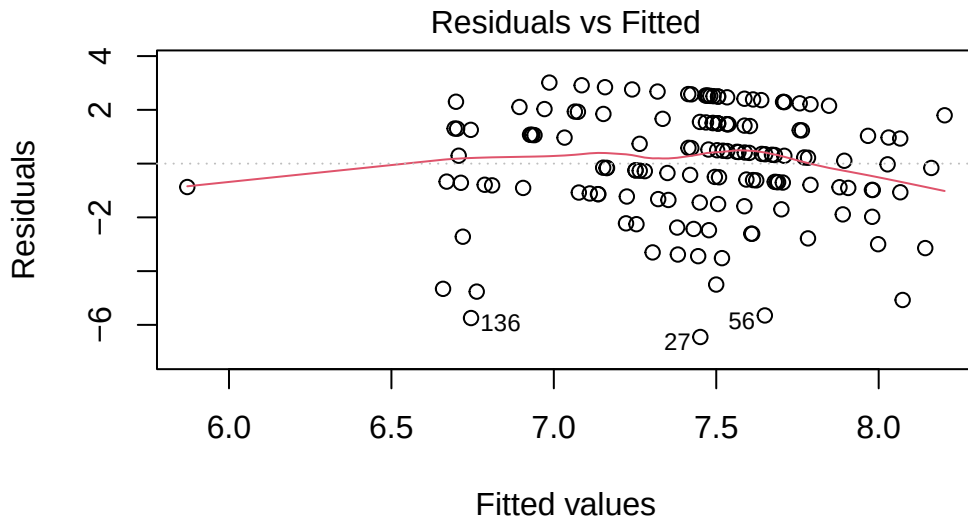


Theoretical Quantiles
 $\text{lm}(\text{'Wake Difficulty Rating'} \sim \text{'Sleep Duration'} + \text{'Sleep Offset Time_Decim'})$

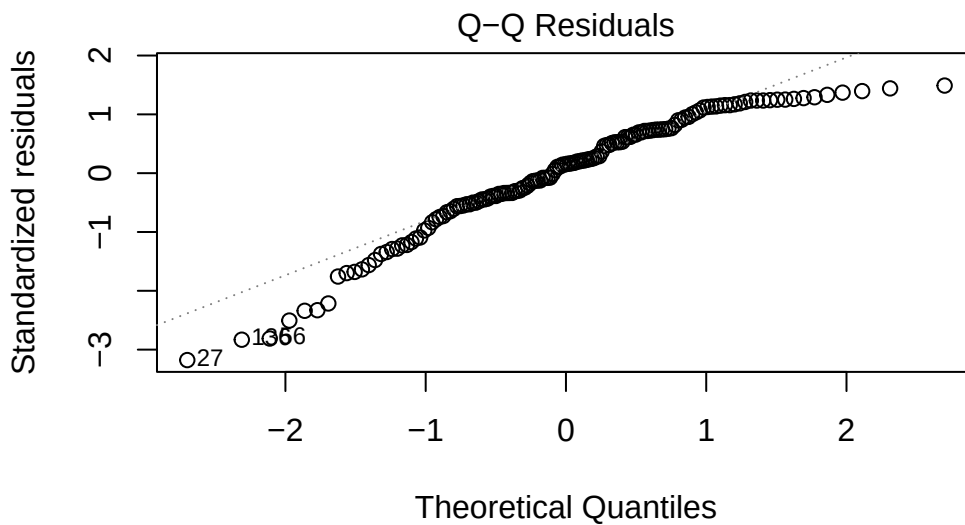




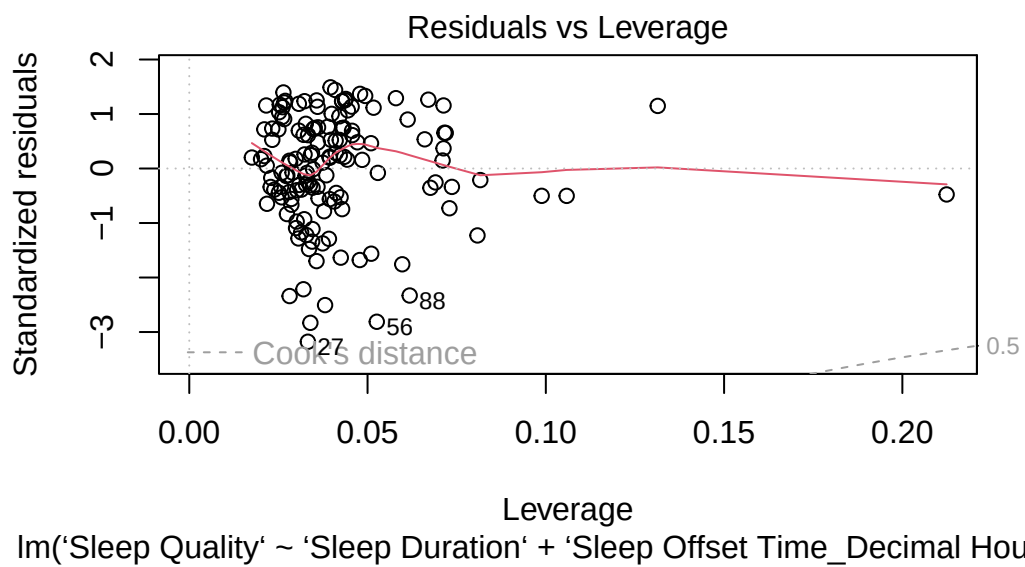
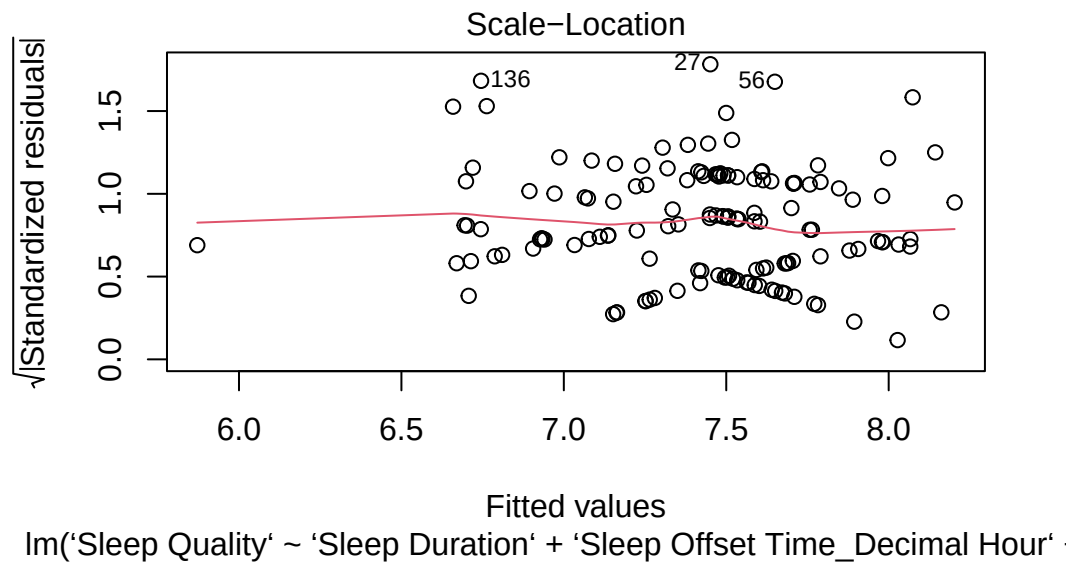


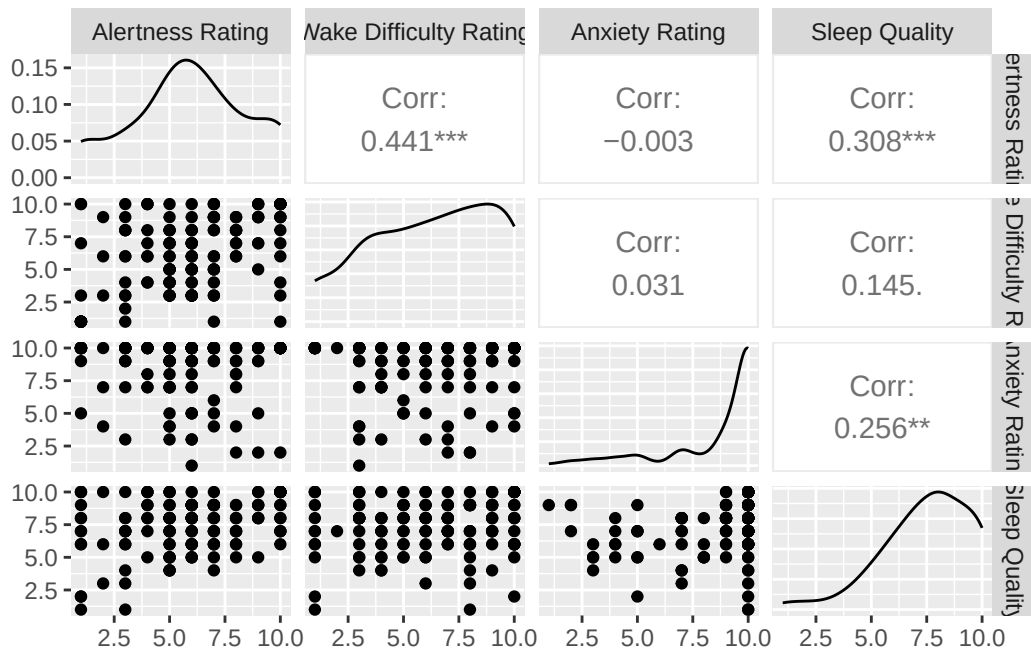
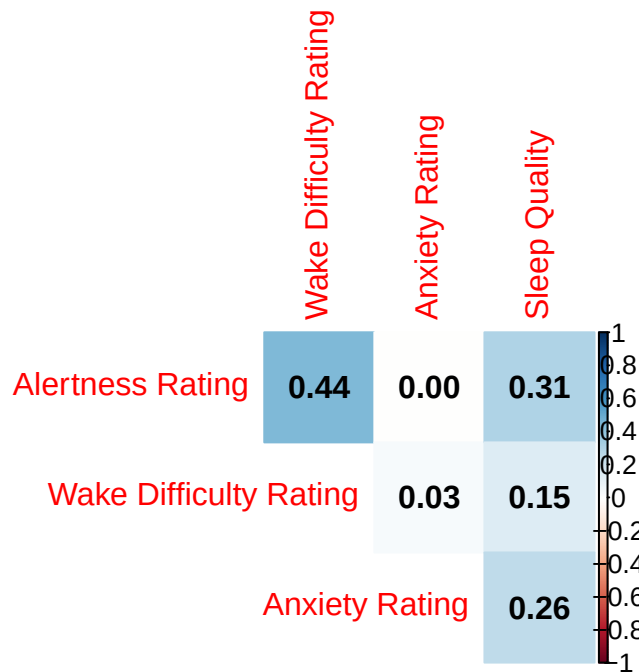


Im('Sleep Quality' ~ 'Sleep Duration' + 'Sleep Offset Time_Decimal Hour')



Im('Sleep Quality' ~ 'Sleep Duration' + 'Sleep Offset Time_Decimal Hour')





t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.71750126	1.20380995	3.0881	0.002433	**
`Sleep Duration`	0.00227718	0.00087143	2.6131	0.009959	**
`Sleep Offset Time_Decimal Hour`	0.03539554	0.13325336	0.2656	0.790921	
`Sleep Onset Decimal Hour`	0.03825066	0.02602504	1.4698	0.143886	
onset_var	0.05858558	0.05074411	1.1545	0.250264	
offset_var	-0.28210566	0.22575268	-1.2496	0.213537	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.99234644	1.32378744	3.7713	0.0002414	***
`Sleep Duration`	0.00112287	0.00087052	1.2899	0.1992770	
`Sleep Offset Time_Decimal Hour`	0.21709630	0.14366658	1.5111	0.1330800	
`Sleep Onset Decimal Hour`	0.03100672	0.02764274	1.1217	0.2639687	
onset_var	-0.06414519	0.05279502	-1.2150	0.2264768	
offset_var	-0.71964942	0.25775204	-2.7920	0.0059928	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	8.35989111	1.67780188	4.9826	1.847e-06	***
`Sleep Duration`	0.00115773	0.00086787	1.3340	0.1844	
`Sleep Offset Time_Decimal Hour`	0.03547316	0.17240382	0.2058	0.8373	
`Sleep Onset Decimal Hour`	-0.02487272	0.02642234	-0.9414	0.3482	
onset_var	-0.03092954	0.04037443	-0.7661	0.4449	
offset_var	-0.05125984	0.32376833	-0.1583	0.8744	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.89216552	1.11874964	5.2667	5.206e-07	***
`Sleep Duration`	0.00025542	0.00070248	0.3636	0.7167	

`Sleep Offset Time_Decimal Hour`	0.13434388	0.11047080	1.2161	0.2260
`Sleep Onset Decimal Hour`	0.03206272	0.02098630	1.5278	0.1289
onset_var	-0.01024923	0.04173218	-0.2456	0.8064
offset_var	-0.03678338	0.18998515	-0.1936	0.8468

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

